

References & Resources

References

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 6. Li, F.; Ackloo, S.; Arrowsmith, C. H.; Ban, F.; Barden, C. J.; et al.
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Parkinson Disease Therapy: Current Strategies and Future Research Priorities. Nature Reviews Neurology (2024).
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9. Taylor, S. S.; Kaila-Sharma, P.; Weng, J.-H.; Aoto, P.; Schmidt, S. H.; Knapp, S.; Mathea, S.; Herberg, F. W.
Kinase Domain Is a Dynamic Hub for Driving LRRK2 Allosterity.

Code & Project Repository

GitHub Repository:

https://github.com/nancyihechu/LRRK2_WD40_Analog_Scoring

Implementation of weighted and SKNN-based scoring for analog prioritization in LRRK2 WD40 drug discovery.

Contact & Networking

LinkedIn:

<https://www.linkedin.com/in/nancy-ihechu/>

Feel free to connect for discussion, collaboration, or feedback.

NANCY IHECHU

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SUMMARY

Computational chemist with experience in molecular modeling, virtual screening, and AI-driven drug discovery. Skilled in free-energy calculations, molecular docking and machine learning, for ligand prioritization and hit identification

EDUCATION

Ph.D. Chemistry (Computational Drug Discovery)
University of Missouri-, USA

Jan.2026 – Present

M.Sc. Chemistry/Chemoinformatics
University of Strasbourg, France (AI in Chemistry)
University of Ljubljana, Slovenia (Chemistry)

Sep.2023 – July.2025

B.Sc. Chemistry
Imo State University, Nigeria

Jan.2015 – Feb.2019

WORK EXPERIENCE

Graduate Research Assistant
Present

May 2025 –

University of Missouri-Columbia, USA

- Secured a competitive \$5,000 research grant to develop quantum-inspired molecular similarity methods for drug discovery
- Developed and benchmarked machine learning models (LightGBM, XGBoost, and Random Forest) to prioritize a library of over 7.7 million compounds for a male contraceptive drug discovery project, improving enrichment of experimentally confirmed PGK2 actives
- Evaluated the binding thermodynamics of over 100 ligand–protein complexes using Absolute Binding Free Energy calculations in GROMACS, supporting hit prioritization for Parkinson's disease drug discovery
- Screened over 10,000 prioritized compounds using molecular docking against PGK2, identifying candidates for downstream experimental validation
- Developed quantum-inspired molecular similarity descriptors by combining feature engineering, RDKit molecular processing, and continuous-time quantum walks to enrich active compounds across the DUDE-Z benchmark
- Reduced molecular dynamics simulation time by 60% through GPU-optimized HPC pipelines, Slurm job scheduling, and large-scale parallel execution

Summer Intern
Novartis Slovenia

March.2025 – April.2025

- Developed automated Python workflows for Host Cell Protein (HCP) data extraction, transformation, visualization, and reporting, contributing to a project aimed at reducing data processing time
- Improved accessibility of HCP datasets by developing interactive dashboards for real-time monitoring of protein data and project progress, supporting cross-functional research teams
- Communicated analytical insights by presenting data-driven findings that supported project planning and operational efficiency

Laboratory Analyst
July.2023

Dec.2019 –

Standards Organization of Nigeria

- Performed chromatographic separation and mass spectrometry analyses on environmental samples, supporting analytical data acquisition, quality assessment, and regulatory laboratory testing

- Improved interpretation of analytical results by developing Python- and Power BI-based workflows that automated quality-control reporting and supported a team of 10 laboratory scientists
- Maintained regulatory compliance and scientific documentation by preparing quality-control records, interpreting analytical results, and ensuring accurate laboratory record-keeping
- Managed laboratory inventories, coordinating equipment maintenance schedules, and ensuring the availability of research supplies

PUBLICATION

Ihechu, N. I. Anti-Parkinson's Therapeutic Discovery: A Computational Approach. Master's Thesis, University of Ljubljana, Faculty of Chemistry and Chemical Technology, 2026 [PID: 20.500.12556/RUL-182772](https://nbn-resolving.org/urn:nbn:de:hbz:5:1-65862-p0011-9)

PRESENTATIONS

American Chemical Society Regional Meeting

Oct. 2025

- Title: Anti-Parkinson's Therapeutic Discovery: A Computational Approach
- Description: Poster Presentation on prioritization workflow in the early phase drug discovery

HONORS AND AWARDS

Erasmus Mundus Double Master's Degree Scholarship

Sep. 2023

- Description: Competitive European Union-funded scholarship awarded for a joint international master's program in Chemistry and Chemoinformatics.

AFFILIATION

- American Chemical Society (ACS) – Member
- Machine Learning Innovation Network For Research to Advance MEDicinal Chemistry (MAINFRAME) – Member